UDACITY
Part of Accenture

Analysis of Salary Data

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Agenda

Findings of linear regression modeling with tech salary data

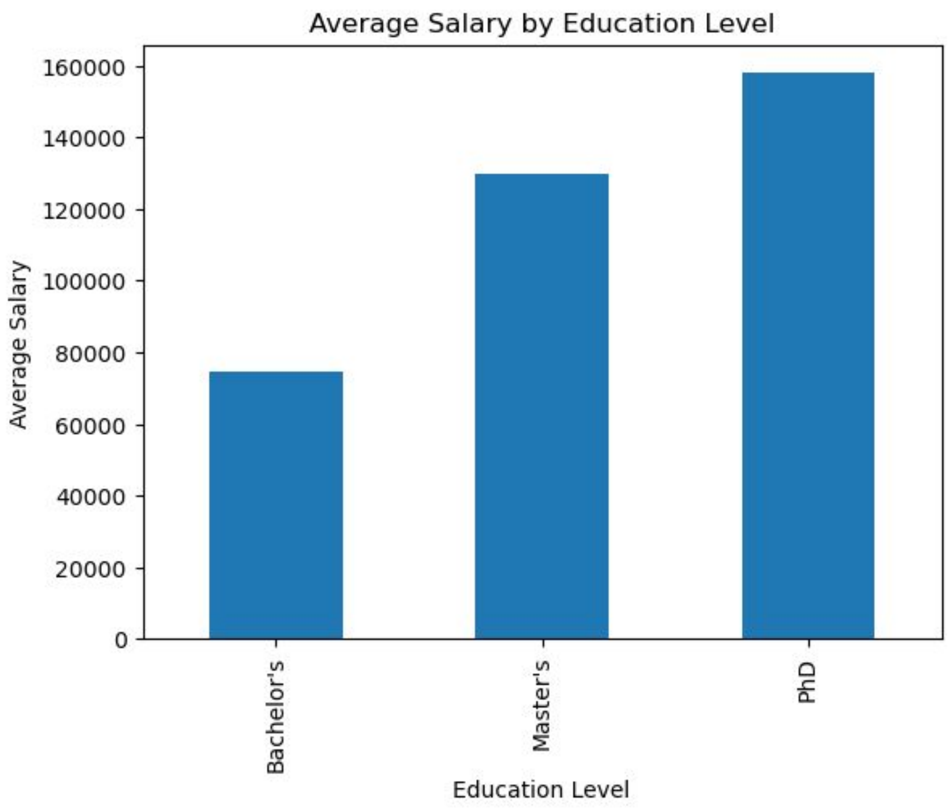
- Data Description
- Regression Results
- Interpretation and Next Steps

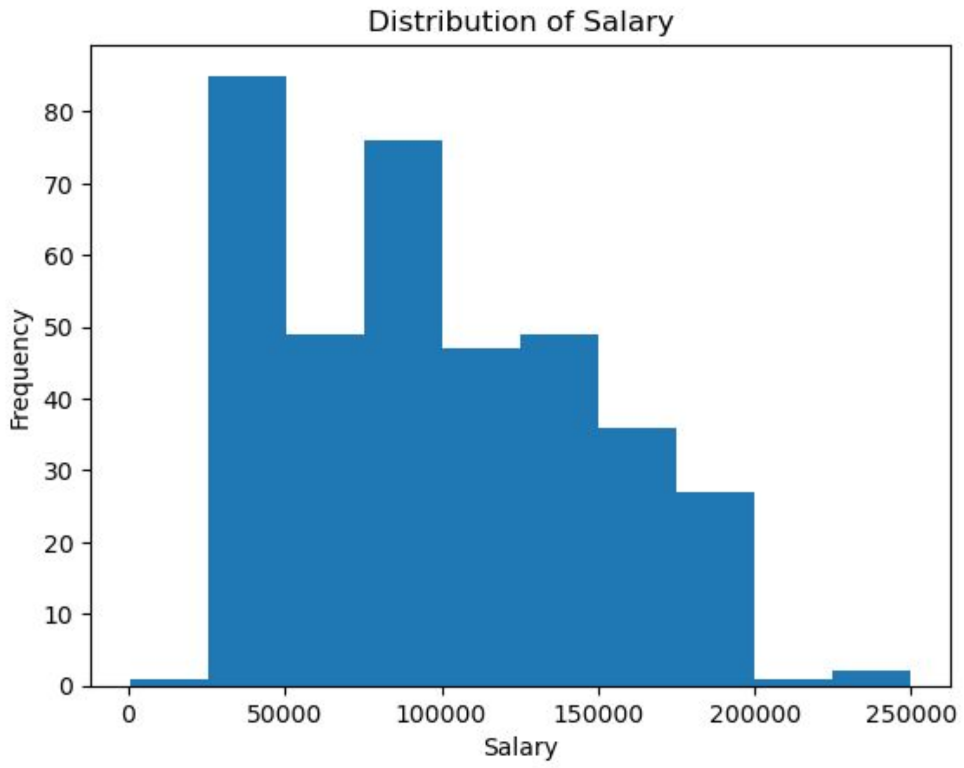
Data Description (1 of 2)

- This dataset originally included 375 employees.
- There was missing data for two employees, which were dropped since they accounted for only a tiny portion, leaving 373 employees.
- Salaries for the 373 employees ranged from $350 to $250,000, with the average salary at $100,577 and the median at $95,000. The histogram for the salary data is right-skewed, where most employee salaries range from around $40,000 to $150,000.
- The following features appear to be correlated with salary:
    - **Education level**: Higher education levels are correlated with higher average salaries: bachelor's ($74,756), master's ($129,796), and Ph.D. ($157,843).
    - **Gender**: Males earn a higher average salary ($103,868) than females ($97,011), a difference of $6,857.
    - **Years of experience**: Average salaries tend to increase year-over-year for professionals with 0 to 25 years of experience; this is also confirmed with the "r" value of 0.93.
    - **Job title**: Having a manager, senior, or director title is correlated with higher salaries.

Data Description (2 of 2)

- Sample of supporting charts





Regression Results (1 of 3)

- Final model output:

OLS Regression Results

Dep. Variable:	Salary	**R-squared:**	0.914
Model:	OLS	**Adj. R-squared:**	0.913
Method:	Least Squares	**F-statistic:**	557.4
Date:	Thu, 18 Jun 2026	**Prob (F-statistic):**	1.46e-190
Time:	19:30:05	**Log-Likelihood:**	-4092.6
No. Observations:	373	**AIC:**	8201.
Df Residuals:	365	**BIC:**	8233.
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	2.876e+04	1677.411	17.146	0.000	2.55e+04	3.21e+04
is_male	7629.6521	1481.834	5.149	0.000	4715.649	1.05e+04
is_senior	1.445e+04	1820.427	7.936	0.000	1.09e+04	1.8e+04
is_manager	4284.6595	1822.582	2.351	0.019	700.579	7868.740
is_director	2.498e+04	3503.028	7.131	0.000	1.81e+04	3.19e+04
Years of Experience	5104.4388	177.226	28.802	0.000	4755.927	5452.950
education_Master's	1.404e+04	2032.078	6.908	0.000	1e+04	1.8e+04
education_PhD	2.308e+04	2760.740	8.360	0.000	1.77e+04	2.85e+04

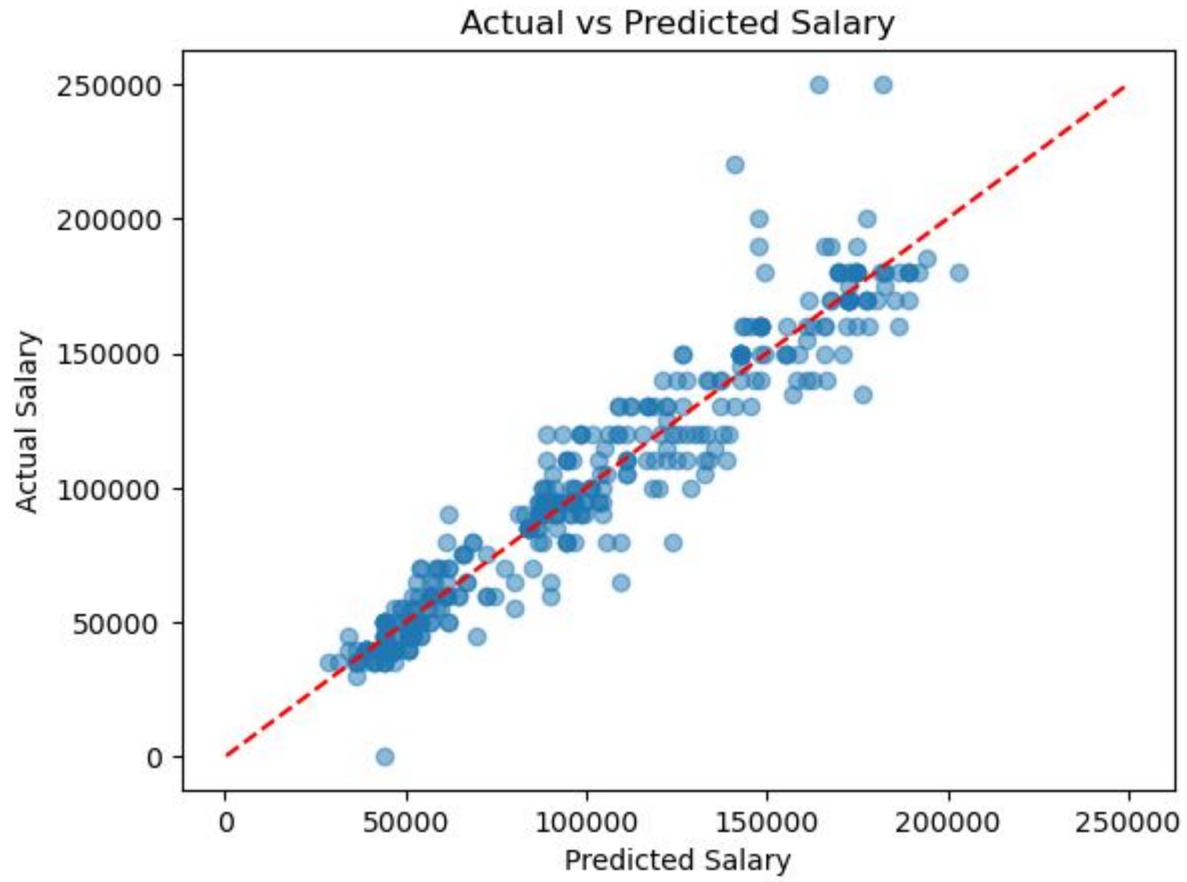
Omnibus:	126.635	**Durbin-Watson:**	1.790
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	1010.895
Skew:	1.196	**Prob(JB):**	3.07e-220
Kurtosis:	10.702	**Cond. No.**	60.8

Regression Results (2 of 3)

- The original model used 10 predictors, with resulting values of 0.915 for R-squared and 0.913 for adjusted R-squared. The final model used 7 predictors with similar resulting values of 0.914 for R-squared and 0.913 for adjusted R-squared.
- These variables were found statistically significant in their relationship with salary, with p<0.05. Their resulting values in the final model are as follows:
    - **is_male (p<0.001, coef=+$7,630)**
    - **is_senior (p<0.001, coef=+$14,450)**
    - **is_manager (p=0.019, coef=+$4,285)**
    - **is_director (p<0.001, coef=+$24,980)**
    - **Years of Experience (p<0.001, coef=+$5,104)**
    - **education_Master's (p<0.001, coef=+$14,040)**
    - **education_PhD (p<0.001, coef=+$23,080)**
- The following variables were found to not be statistically significant based on the full model results and were removed when creating the final model:
    - **is_junior (p=0.068, coef=-$4,493)**
    - **is_analyst (p=0.933, coef=-$176)**
    - **is_engineer (p=0.508, coef=+$2,363)**

Regression Results (3 of 3)

- The model appears to fit the data relatively well based on the R-squared and adjusted R-squared values.
- We can conclude that the model accounts for around 91% of the variance in salary.
- A scatterplot of the actual versus predicted salary data confirms a good fit, despite some outliers, especially for higher salaries.



Interpretation and Next Steps

- **Variable 1: Senior title (compared to someone with no other title indications)**
  - Given that all other variables remain constant, salary is expected to increase by approximately $14,450 for an employee with a senior title.
  - I'm 95% confident that the difference in salary will be around $10,900 to $18,000.
- **Variable 2: Years of experience**
  - With each additional year of experience, salary is expected to increase by approximately $5,104.
  - I'm 95% confident the salary will increase by approximately $4,756 and $5,453 for each additional year of experience.
  - **Potential next steps:**
    - Consider further cleaning the data to address outliers that can skew the model.
    - Add more features to see whether they might impact salary, such as the employee's location, age, company size, and race.
    - Experiment with machine learning and scikit-learn for predicting new salary data.